(b) Matrix multiplication is distributive over matrix addition

(i) If $A,\ B,\ C$ are $m\times n$, $n\times p$ and $n\times p$ matrices respectively then $A(B+C)=AB+AC$ (Right Distributive Property)

(ii) If $A,\ B,\ C$ are $m\times n$, $m\times n$ and $n\times p$ matrices respectively then $(A+B)C=AC+BC$ (Left Distributive Property)

(c) Matrix multiplication is always associative

If $A,\ B,\ C$ are $m\times n$, $n\times p$ and $p\times q$ matrices respectively then $(AB)C=A(BC)$

(d) Multiplication of a matrix by a unit matrix

If A is a square matrix of order $n\times n$ and I is the unit matrix of same order then $AI=IA=A$.

Note

- If x and y are two real numbers such that $xy=0$ then either $x=0$ or $y=0$. But this condition may not be true with respect to two matrices.
- $AB=0$ does not necessarily imply that $A=0$ or $B=0$ or both $A,B=0$

Illustration

$$A=\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}\neq 0 \quad \text{and} \quad B=\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\neq 0$$

But $$AB=\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}\times\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}=\begin{pmatrix}1-1 & 1-1\\ -1+1 & -1+1\end{pmatrix}=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}=0$$

Thus $A\neq 0,\ B\neq 0$ but $AB=0$.

Example 3.67 If $A=\begin{pmatrix}1 & 2 & 0\\ 3 & 1 & 5\end{pmatrix}$, $B=\begin{pmatrix}8 & 3 & 1\\ 2 & 4 & 1\\ 5 & 3 & 1\end{pmatrix}$, find AB.

Solution We observe that A is a 2×3 matrix and B is a 3×3 matrix, hence AB is defined and it will be of the order 2×3.

Given $A=\begin{pmatrix}1 & 2 & 0\\ 3 & 1 & 5\end{pmatrix}_{2\times 3}$, $B=\begin{pmatrix}8 & 3 & 1\\ 2 & 4 & 1\\ 5 & 3 & 1\end{pmatrix}_{3\times 3}$

$$AB=\begin{pmatrix}1 & 2 & 0\\ 3 & 1 & 5\end{pmatrix}\times\begin{pmatrix}8 & 3 & 1\\ 2 & 4 & 1\\ 5 & 3 & 1\end{pmatrix}$$

$$=\begin{pmatrix}8+4+0 & 3+8+0 & 1+2+0\\ 24+2+25 & 9+4+15 & 3+1+5\end{pmatrix}=\begin{pmatrix}12 & 11 & 3\\ 51 & 28 & 9\end{pmatrix}$$

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